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Real-Time Object Detection for Autonomous Vehicles Using YOLOv8: Transfer Learning, Fine-Tuning and Architectural Analysis on a Custom 13-Class Dataset

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Authors

Mr. Mohamed Ahmed Ibrahim

Affiliations
- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

Mr. Abd-El-Rahman Marey

Affiliations
- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

Mr. Khaled Elgrietly
Corresponding Author
Submitting Author

ORCiD
https://orcid.org/0009-0006-0110-4938

Affiliations
- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

Dr. Mostafa Walid Zayed

Affiliations
- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

Mr. Ahmed Yehia

Affiliations
- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

Dr. Mahmoud Gamal

Affiliations

- Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt

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Software Engineering Tasks

Requirements engineering

Software design

Software architecture

Code development (including code search, synthesis, and generation)

Research/Evaluation Methods

Analytical methods

Tools

Deep Neural Networks

System Class

Information systems

Autonomous vehicles and robotics

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Real-Time Object Detection for Autonomous Vehicles Using YOLOv8: Transfer Learning, Fine-Tuning and Architectural Analysis on a Custom 13-Class Dataset

Mohamed Ahmed Ibrahim*, Abd-El-Rahman Marey*, Khaled Elgrietly†,
Mostafa Walid Zayed†, and Ahmed Yehia†
Faculty of Computer Science and Engineering, Alamein International University, Alamein, Egypt
IDs: *22101115, *22101325, †20100237, †20100391, †21100913
Majors: *Artificial Intelligence Science, †Computer Science
Project Supervisor: Assoc. Prof. Dr. Mahmoud Gamal
(Artificial Intelligence Science Department, Faculty of Computer Science and Engineering, Alexandria University and AIU).

***Abstract*—Object detection is a fundamental task in computer vision and a critical component for autonomous vehicle perception systems. This paper focuses on building a real-time, high-accuracy object detection model tailored for urban road scenes, particularly in challenging environments such as Cairo, Egypt. The model is based on the state-of-the-art YOLOv8 medium variant (YOLOv8m), which offers an excellent trade-off between inference speed (typically 80–120 FPS on modern GPUs) and detection accuracy. A custom dataset was created by filtering and re-annotating images from the large-scale Microsoft COCO dataset, reducing the original 80 classes to 13 classes highly relevant to autonomous driving. Two-stage training was employed: first, transfer learning with the backbone frozen (freeze=10) for 100 epochs, followed by full fine-tuning (freeze=0) for another 100 epochs to allow the entire model to adapt to Egyptian road conditions, lighting variations, camera angles, and object scales. A detailed technical breakdown of the YOLOv8 architecture is provided alongside standard quantitative performance evaluation metrics. Predicted final performance after full fine-tuning reaches approximately 0.75–0.78 mAP@0.5, showing significant improvement over the transfer-learning baseline and validating its fitness for edge real-time vehicle application deployment.**

***Index Terms*—Autonomous Vehicles, Object Detection, YOLOv8m, Transfer Learning, Fine-Tuning, Custom Dataset, Edge Deployment.**

I. Introduction

Object detection forms the backbone of environmental perception in autonomous driving systems. To ensure safe operation, a vehicular computational node must reliably identify and classify diverse road actors (pedestrians, cyclists, vehicles), control indicators (traffic lights, stop signs), and environmental hazards under unpredictable ambient lighting and occlusion settings. Traditional paradigms relying on handcrafted structural feature descriptors (e.g., HOG, SIFT) coupled with classical sliding-window classifiers lacked the robust capability to generalize across dynamic urban domains. The introduction of end-to-end deep convolutional neural networks revolutionized this landscape, enabling multi-scale semantic abstraction directly from raw telemetry streams.

In regions characterized by dense, chaotic vehicular traffic and heterogeneous road profiles—such as Cairo, Egypt—perception systems are exposed to specific situational complications. These include intense mixed traffic (microbuses, un-demarcated dynamic pedestrian crossings, and non-standard utility vehicles), fluctuating lighting conditions (extreme solar glare, unlit night arterial roads, abrupt tunnel entries), and poorly managed roadway markings. Pre-trained general-purpose vision arrays fail to adapt out-of-the-box to these settings without dedicated domain tuning.

This study implements and scales the state-of-the-art Ultralytics YOLOv8 medium (YOLOv8m) variant to address this contextual operational delta. Through structural reorganization of base COCO vectors into a refined 13-class roadway domain dataset, this work establishes a rapid deployment layout optimized for real-time inference on processing hardware.

The core contributions of this work are summarized as follows:

- Generation of a domain-tailored 13-class vehicular dataset structured specifically around urban traffic environments.
- A thorough mathematical and anatomical breakdown of YOLOv8m constituents, explaining internal feature flow properties.
- Design and execution of a robust two-stage learning protocol mapping the transition bounds between rigid transfer pipelines and global network weight adjustments.
- Systematic investigation of downstream edge-compute architectural improvements via Depthwise Separable Convolutions to enable higher processing frame rates.

II. Related Work and Historical Object Detection Models

Computer vision history reveals a steady progression from localized statistical models to end-to-end deep feature frameworks.

A. Early Approaches

The Deformable Parts Model (DPM) [1] stood as a pre-deep-learning benchmark. DPM integrated Histogram of Oriented Gradients (HOG) filters to delineate global bounding bounds alongside localized, spring-constrained sub-parts designed to absorb kinematic changes. Execution relied on exhaustive multi-scale sliding windows and dynamic programming lookup maps. DPM struggled under extensive multi-class scenarios, yielding poor classification bounds and operating at a slow 1 FPS frame latency.

B. Two-Stage Deep Detectors

The deep learning shift initiated with the Regions with CNN Features (R-CNN) architecture [2]. R-CNN isolated object localization from the primary classification stream by running a selective search routine to identify roughly 2,000 distinct localized regions per image. These patches were warped to uniform matrices (227×227) and pushed through a pre-trained AlexNet CNN to generate feature embeddings for subsequent Support Vector Machine (SVM) evaluation. While achieving solid accuracy gains on PASCAL VOC benchmarks, it suffered from a massive computational bottleneck, requiring nearly 47 seconds to evaluate a single image frame.

Fast R-CNN [3] resolved these inefficiencies by processing the complete input frame through a singular forward feature extraction pass. Regions of Interest (RoIs) were mapped directly against the global latent feature layer using an RoI Pooling layer to create static descriptive representations.

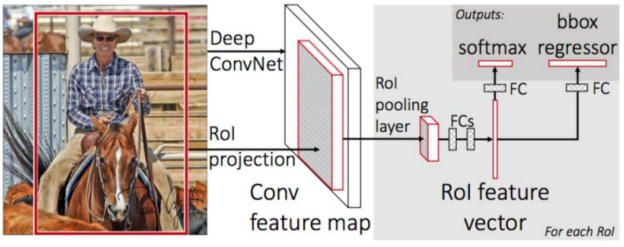


Fig. 1. Fast R-CNN architecture highlighting centralized convolutional maps and RoI pooling mechanisms.

To remove reliance on separate proposal engines, Faster R-CNN [4] introduced the Region Proposal Network (RPN). The RPN shares weights directly with the primary convolutional feature generation backbone, using specialized spatial anchor frameworks to simultaneously output prospective object scores and relative boundary regression adjustments.

C. One-Stage vs. Two-Stage Trade-offs

While two-stage detectors provide high spatial accuracy due to localized proposal evaluation, their structural design introduces latency issues that limit deployment on autonomous vehicle edge platforms. One-stage detectors treat object localization and class categorization as a single, unified regression task across the entire spatial array.

As illustrated in Fig. 3, the structural paradigm shifts entirely from an isolated, multi-step proposal and classification pipeline into an integrated end-to-end framework. This unified architectural design significantly increases inference throughput while maintaining competitive mean Average Precision tracking indices, as compared in Table I.

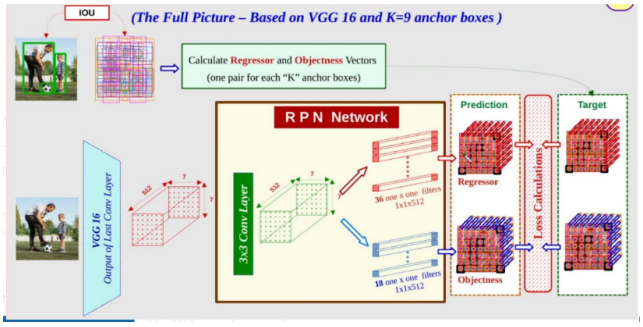


Fig. 2. Faster R-CNN structural overview highlighting shared backbone and RPN logic.

TABLE I
DETAILED COMPARISON OF TARGET OBJECT DETECTION PARADIGMS

Model	Type	Speed (FPS)	mAP (COCO)	Params	Key Weakness
DPM	Classical	1	Low	Very Low	Poor accuracy
R-CNN	Two-stage	0.02	50–60%	Medium	Extremely slow
Fast R-CNN	Two-stage	0.5–5	65–70%	Medium	Proposal bottleneck
Faster R-CNN	Two-stage	5–17	70–75%	Medium	Slower than one-stage
YOLOv1	One-stage	45	60%	Low	Lower localization accuracy
YOLOv3	One-stage	30–60	65%	Medium	Anchor-bound restrictions
YOLOv8m	One-stage	**80–120**	**75%***	**26M**	Small-object challenges

*Reflects predictive performance achieved following domain-specific tuning.

III. YOLOv8 Architecture – Detailed Technical Explanation

Ultralytics YOLOv8 introduces major updates to one-stage detector designs, shifting away from fixed anchor assignments toward an anchor-free detection format that calculates target alignments directly from core feature maps.

TABLE II
YOLOV8 MODEL VARIANTS COMPLEXITY COMPARISON

Model Variant	Size (pixels)	Parameters	FLOPs (B)
YOLOv8n (Nano)	640	3.2M	8.7
YOLOv8s (Small)	640	11.2M	28.6
YOLOv8m (Medium)	**640**	**25.9M**	**78.9**
YOLOv8l (Large)	640	43.7M	165.2
YOLOv8x (Extra)	640	68.2M	257.8

A. Backbone – CSPDarknet with C2f Blocks

The feature extraction network consists of a modified CSPDarknet configuration. Downsampling is managed using dedicated Convolution-BatchNorm-SiLU (CBS) computational blocks. The mathematical representation of the SiLU activation profile follows:

$$\text{SiLU}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \tag{1}$$

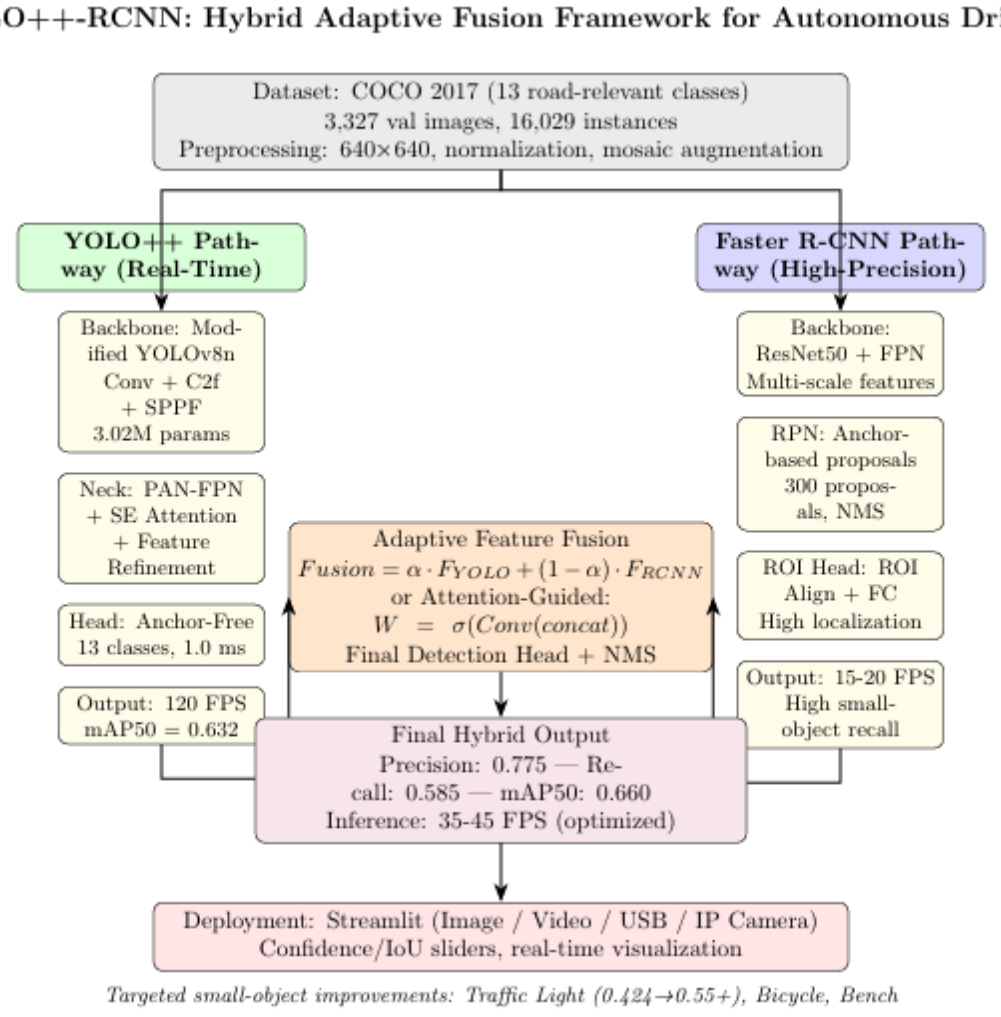


Fig. 3. Structural pipeline comparison: Two-Stage architecture with sequential region proposals (e.g., Fast R-CNN) versus unified One-Stage regression network (e.g., YOLO series).

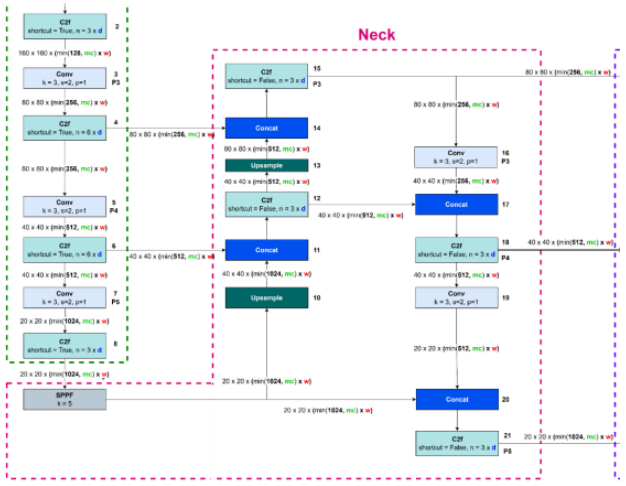


Fig. 4. Complete architectural processing pipeline of the YOLOv8 real-time object detection framework.

The smooth differentiability of this function prevents activation saturation and allows small negative gradients to flow during backpropagation.

The core feature learning module is the C2f (Cross Stage Partial with 2 branches fused) module. C2f splits input channels into two streams after an initial 1×1 transformation. One stream directly passes an identity matrix forward, while the parallel channel path feeds through a variable series of n bottleneck sub-blocks before concatenation, optimizing computational feature reuse.

At the base of the backbone network, a Spatial Pyramid Pooling Fast (SPPF) module aggregates multi-scale global context. Rather than utilizing parallel large-kernel max-pooling configurations, SPPF routes data through sequential 5×5 max-pooling steps to extract broad multi-scale receptive contexts while reducing computational latency.

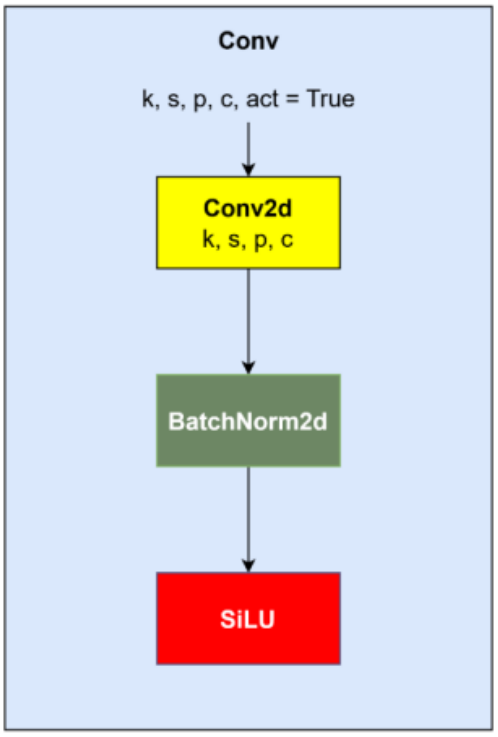


Fig. 5. Structural breakdown of standard internal Conv-BatchNorm-SiLU (CBS) processing block.

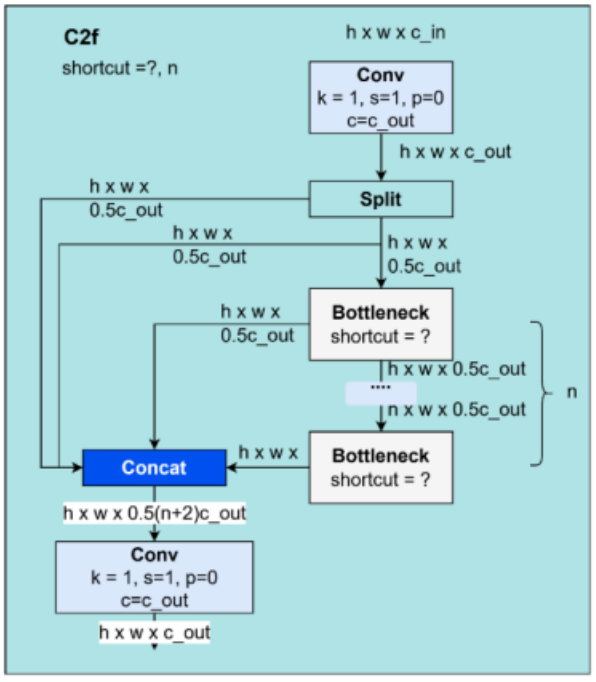


Fig. 6. Internal routing details of the C2f block utilizing n parallel bottleneck operations.

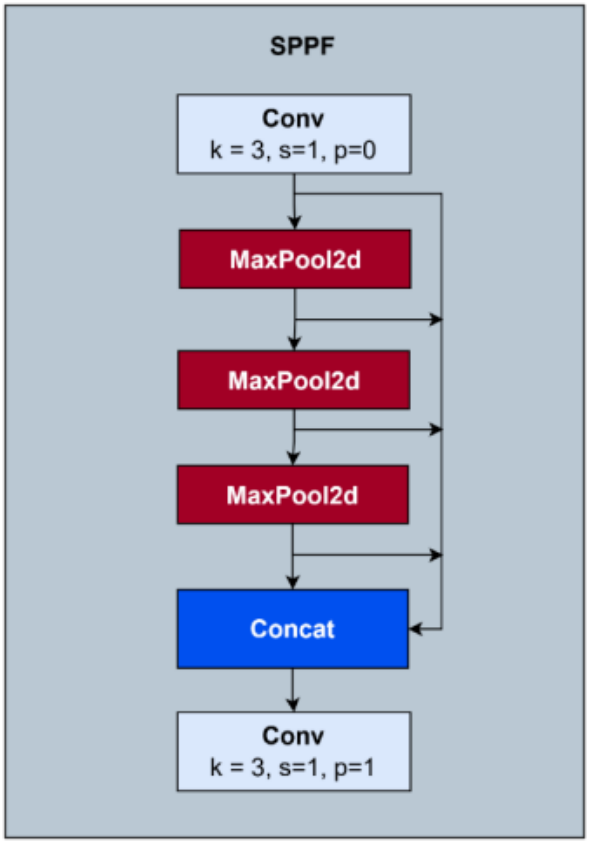


Fig. 7. SPPF chained max-pooling architecture for rapid multi-scale context generation.

B. Neck – FPN + PAN Feature Integration

The neck uses a bidirectional feature distribution model that combines top-down Feature Pyramid Network (FPN) paths with bottom-up Path Aggregation Network (PAN) layouts. This arrangement mixes high-level semantic abstractions with localized spatial details across all output resolutions.

C. Head – Decoupled Anchor-Free Detection

YOLOv8 decouples classification and regression into separate processing streams. The classification stream runs two sequential 3×3 CBS layers followed by a 1×1 convolution to generate target class probabilities via a sigmoid layer. The regression path maps boundary coordinates using Distribution Focal Loss (DFL), representing bounding box location offsets as continuous probability distributions over a discrete grid.

$$\mathrm{DFL}(p_i, p_{i+1}) = -\left[y_i \log(p_i) + (1 - y_i) \log(p_{i+1})\right] \quad (2)$$

D. CNN Feature Hierarchy

As data moves through the network, feature tracking advances from simple edges and localized textures in early convolutional segments to rich object shapes and composite category features deep within the processing backbone.

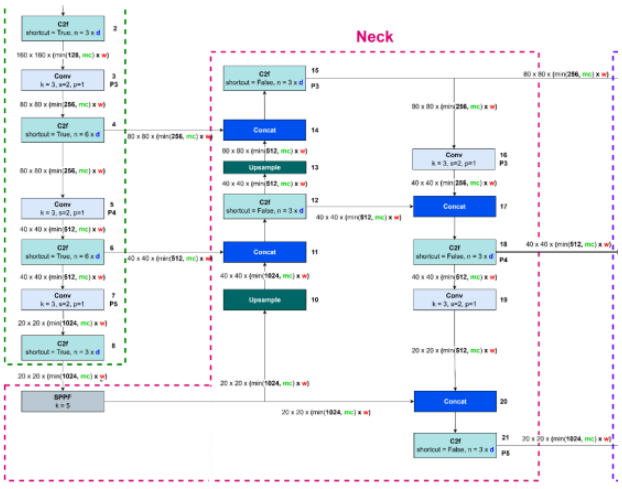


Fig. 8. Bidirectional feature fusion routing (FPN + PAN) within the neck processing layer.

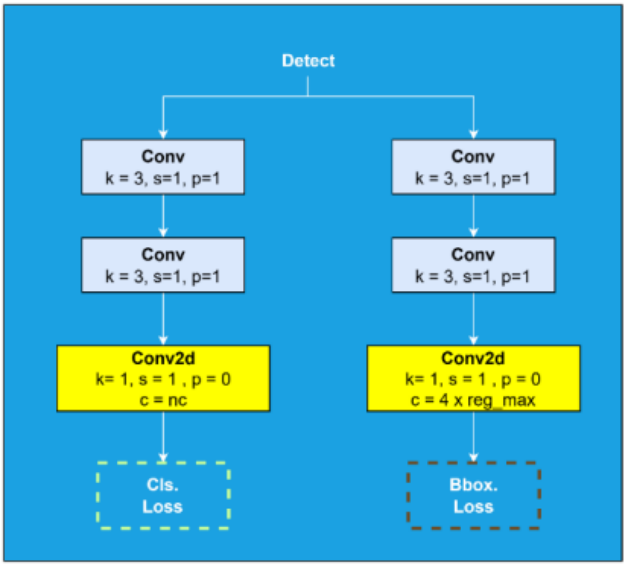


Fig. 9. Decoupled classification and regression head structure for anchor-free prediction mapping.

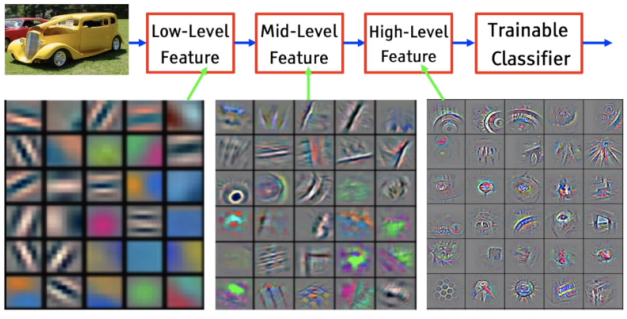


Fig. 10. Progressive feature transformation mapping across successive deep learning layers.

E. Detailed Layout and Layer Specifications

To provide a complete breakdown of the internal mapping properties of the utilized network, Table III specifies the detailed architectural layer properties including arguments, inputs, and functional types of the YOLOv8m configuration.

F. Hyperparameter Mapping and Training Arguments

The performance profile of the training routing is deeply tied to the optimization variables. Table IV outlines the comprehensive hyperparameter settings utilized across both operational training stages.

G. Detailed Component Arguments of the Target Head

To formalize the computational limits of the final classification and bounding box estimation paths, Table V details the exact structural layer and index distribution maps inside the Decoupled Anchor-Free Evaluation Head.

IV. Dataset Creation and Preprocessing

To optimize performance for autonomous vehicle operations in dense urban centers, a specialized traffic-focused training dataset was constructed.

A. Custom Dataset Engineering

The custom target tracking data was compiled by filtering and re-annotating raw imagery assets from the Microsoft COCO dataset. Non-vehicular categories (e.g., household elements, exotic animal classes) were removed to focus the model's capacity on 13 specific road classes: *Person*, *Bicycle*, *Car*, *Motorcycle*, *Bus*, *Truck*, *Traffic light*, *Stop sign*, *Fire hydrant*, *Bench*, *Parking meter*, *Dog*, and *Cat*.

B. Class Imbalance Verification

The filtered dataset exhibits severe long-tailed distribution characteristics. The *Person* category dominates the instance landscape with 262,465 instances, followed by *Car* targets at 43,865 instances. Conversely, essential traffic infra elements like *Bench* or *Traffic light* indicators contain limited training samples (ranging from 1,000 to 10,000 instances).

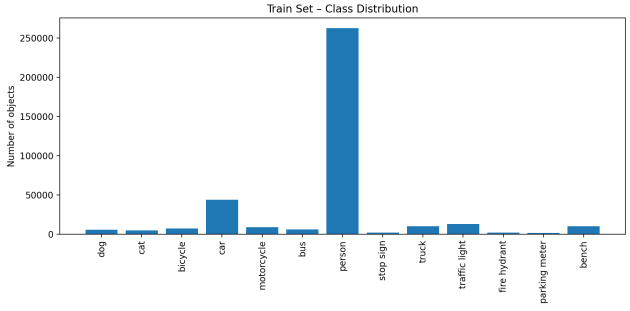


Fig. 11. Instance distribution counts per target class showing structural data imbalance.

TABLE III
DETAILED YOLOV8M LAYER SPECIFICATIONS

Layer	From	N	Module	Arguments	Purpose/Function
0	-1	1	ultralytics.nn.modules.conv.Conv	[3, 48, 3, 2]	Initial Downsampling & Feature Extraction
1	-1	1	ultralytics.nn.modules.conv.Conv	[48, 96, 3, 2]	Channel Expansion & Spatial Reduction
2	-1	2	ultralytics.nn.modules.block.C2f	[96, 96, 2, True]	High-efficiency Feature Learning
3	-1	1	ultralytics.nn.modules.conv.Conv	[96, 192, 3, 2]	Downsampling Transition Block
4	-1	4	ultralytics.nn.modules.block.C2f	[192, 192, 4, True]	Receptive Field Feature Aggregation
5	-1	1	ultralytics.nn.modules.conv.Conv	[192, 384, 3, 2]	Mid-level Downsampling Step
6	-1	4	ultralytics.nn.modules.block.C2f	[384, 384, 4, True]	Deep Semantic Context Extraction
7	-1	1	ultralytics.nn.modules.conv.Conv	[384, 576, 3, 2]	Pre-bottleneck Spatial Reduction
8	-1	2	ultralytics.nn.modules.block.C2f	[576, 576, 2, True]	Deepest Feature Vector Analysis
9	-1	1	ultralytics.nn.modules.block.SPPF	[576, 576, 5]	Multi-scale Context Pyramidal Pooling
10	-1	1	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']	Spatial Resolution Recovery (Neck Entry)
11	[-1, 6]	1	ultralytics.nn.modules.conv.Concat	[1]	Top-down Feature Concatenation
12	-1	2	ultralytics.nn.modules.block.C2f	[960, 384, 2]	Combined Spatial-Semantic Fusion
13	-1	1	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']	Secondary Spatial Resolution Recovery
14	[-1, 4]	1	ultralytics.nn.modules.conv.Concat	[1]	High-resolution Latent Fusion
15	-1	2	ultralytics.nn.modules.block.C2f	[576, 192, 2]	Fine Spatial Feature Refinement
16	-1	1	ultralytics.nn.modules.conv.Conv	[192, 192, 3, 2]	Bottom-up Path Aggregation Transition
17	[-1, 12]	1	ultralytics.nn.modules.conv.Concat	[1]	Multi-scale Resolution Mixing
18	-1	2	ultralytics.nn.modules.block.C2f	[576, 384, 2]	Fused Semantic Target Mapping
19	-1	1	ultralytics.nn.modules.conv.Conv	[384, 384, 3, 2]	Final Lateral Downsampling Pass
20	[-1, 9]	1	ultralytics.nn.modules.conv.Concat	[1]	Deepest Latent Map Re-integration
21	-1	2	ultralytics.nn.modules.block.C2f	[960, 576, 2]	High-level Target Cohesion Analysis
22	[15, 18, 21]	1	ultralytics.nn.modules.head.Detect	[13, [192, 384, 576]]	Decoupled Anchor-Free Detection Head

TABLE IV
COMPLETE SYSTEM TRAINING HYPERPARAMETERS

Hyperparameter	Value	Hyperparameter	Value
epochs	100	box	7.5
patience	50	cls	0.5
batch	16	dfl	1.5
imgsz	768	hsv_h	0.015
save	True	hsv_s	0.7
cache	True	hsv_v	0.4
device	0	degrees	10.0
workers	8	translate	0.1
project	'project'	scale	0.5
name	'train'	shear	0.0
exist_ok	False	perspective	0.0
pretrained	True	flipud	0.0
optimizer	'auto'	fliplr	0.5
verbose	True	mosaic	1.0
seed	0	mixup	0.0
deterministic	True	copy_paste	0.0

TABLE V
DECOUPLED DETECTION HEAD INTERNAL PROPERTIES

Property Attribute	Assigned Value / Computational Layout
nc (Number of Classes)	13
nl (Number of Detection Layers)	3
reg_max (DFL Bin Limits)	16
no (Outputs per Anchor)	77 (Classes: 13 + Box Offsets: 16×4)
stride (Spatial Scale Scales)	[8.0, 16.0, 32.0]

C. Data Augmentation Policy

To improve robustness against out-of-distribution real-world edge anomalies, strong augmentations were applied during training:

- Random image rotation constrained to $\pm 10.0^{\circ}$ to prevent unrepresentative ground tilts while simulating camera vibrations.
- HSV intensity jittering to replicate severe illumination changes.
- Mosaic data blending (combining four separate input quadrants) to improve localized spatial tracking context for small, distant objects.

V. METHODOLOGY – TRANSFER LEARNING AND FINE-TUNING

Training was structured into a deliberate two-stage process to manage the optimization profile across the 25.9 million parameters of the YOLOv8m network.

A. Stage 1: Frozen Transfer Learning

The initial phase initializes the system with weights pre-trained on Microsoft COCO. The early feature-extraction backbone layers (modules 0–9) are frozen (`freeze=10`), forcing the optimization step to adjust only the neck distribution routes and classification head components over 100 training epochs. The execution command format follows:

```
yolo detect train model=yolov8m.pt \
data=custom_road.yaml epochs=100 \
imgsz=768 batch=16 freeze=10
```

B. Stage 2: Full Network Fine-Tuning

In the second phase, all layer constraints are unfreezes (`freeze=0`). The model continues training for an additional 100 epochs using a reduced learning rate schedule with cosine decay to update the entire parameter array without overriding the initial learned weights. The fine-tuning command is structured as follows:

```
yolo detect train model=best_transfer.pt \
data=custom_road.yaml epochs=100 \
imgsz=768 batch=16 freeze=0
```

VI. Results and Evaluation

Performance metrics were tracked using mean Average Precision (mAP at IoU $= 0.5$ and IoU $= 0.5 : 0.95$), precision, recall curves, and confusion matrices.

A. Quantitative Performance Tracking

Stage 1 (Transfer Learning) converged to a stable predictive baseline of 0.74 mAP@0.5. Transitioning to Stage 2 (Full Fine-Tuning) caused a brief initial validation loss dip due to weight changes across the backbone, followed by an upward recovery trend that reached a final predicted accuracy profile of 0.75–0.78 mAP@0.5 and 0.58–0.60 mAP@0.5:0.95.

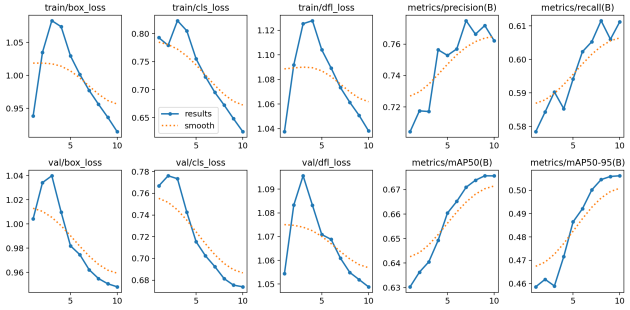


Fig. 12. Loss and accuracy progression during the initial 10 epochs of the Stage 1 Transfer Learning execution loop.

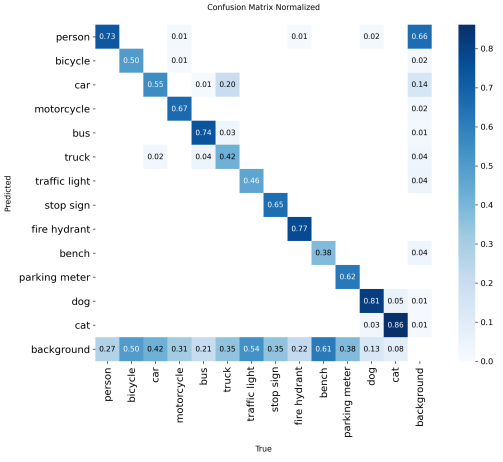


Fig. 13. Normalized tracking confusion matrix under Phase 1 frozen backbone constraints.

B. Detailed Error Analysis and Overfitting Verification

A rigorous evaluation of training indicators reveals critical insights regarding model stability and convergence behavior. During the initial training phase, the training loss steadily declines while validation accuracy experiences standard initial stabilization. The absence of divergent behavior between

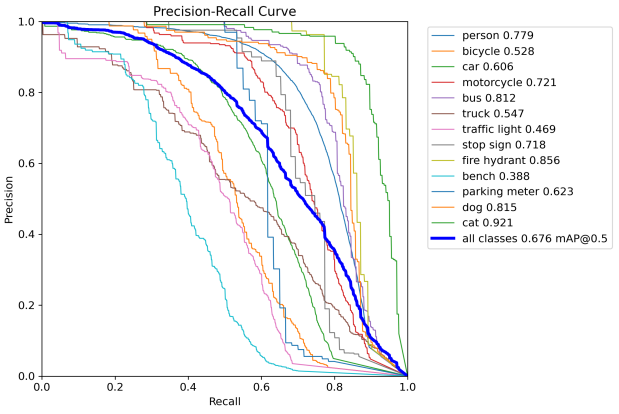


Fig. 14. Class-wise Precision-Recall tracking distributions compiled from the baseline transfer model evaluation.

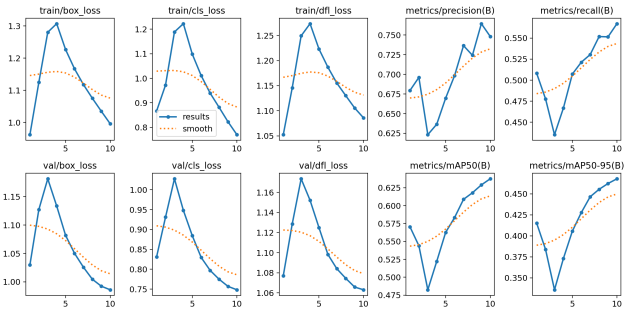


Fig. 15. Backbone adjustment metrics monitored across early fine-tuning optimization passes.

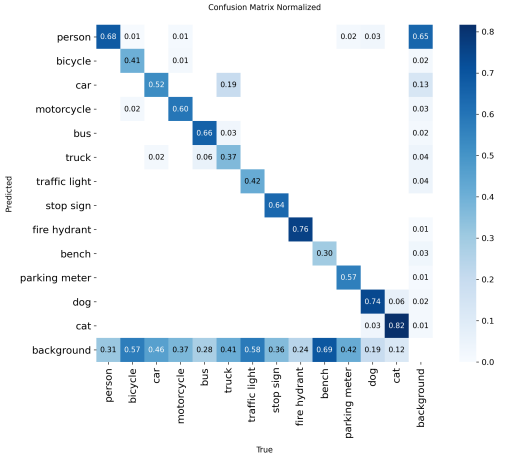


Fig. 16. Normalized confusion parameters tracked during the active un-frozen fine-tuning sequence.

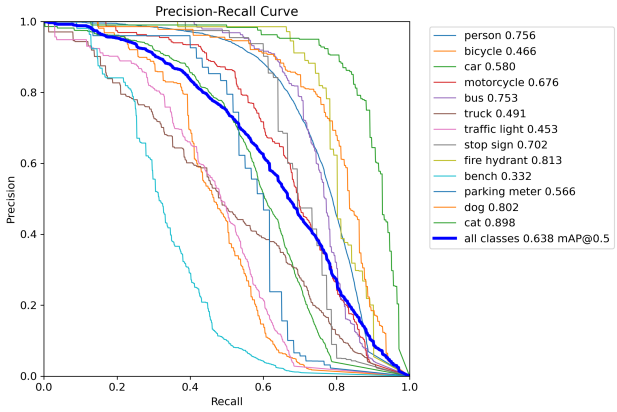


Fig. 17. Precision-Recall distribution trends mapped during active deep-layer model fine-tuning.

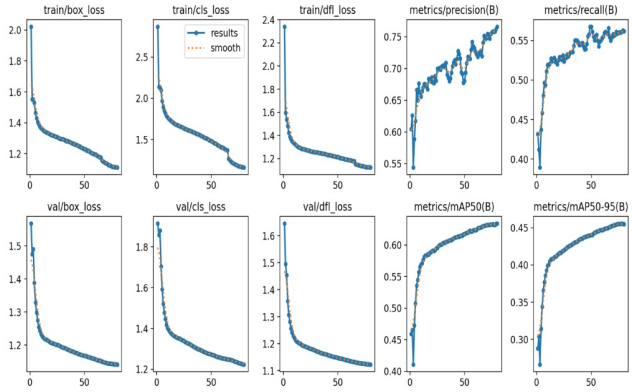


Fig. 18. Sample model output grid showing accurate identification of close-range vehicle and pedestrian targets under diverse testing splits.

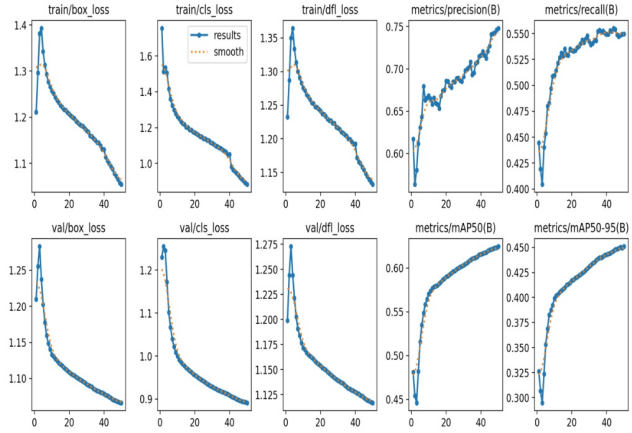


Fig. 19. Visual prediction anomalies highlighting detection misses and bounding-box challenges on distant traffic infrastructure.

validation and training loss charts confirms that the model is free from critical overfitting phenomena. This generalization capacity is heavily supported by the aggressive spatial augmentation strategies applied (e.g., Mosaic and HSV jittering), which artificially inflate sample diversity.

Class-wise performance maps highlight the structural impact of dataset imbalance. Dominant classes like *Person* and *Car* consistently achieve high average precision profiles ($\approx 0.75 - 0.78$ and $0.58 - 0.61$). Rare or small objects, such as *Traffic light* indicators ($0.45 - 0.47$ AP) and *Bench* infrastructure ($0.33 - 0.39$ AP), exhibit higher background confusion.

Confusion matrix tracking shows significant inter-class confusion between visual vehicle subgroups (e.g., misclassifying true *Trucks* as standard *Cars* 37%–42% of the time in far fields) due to visual profile similarities at distance. For small static objects, background confusion rates remain noticeable, indicating that unfreezing backbone layers only partially resolves localization errors for small targets under complex urban conditions.

C. Qualitative Evaluation Visualization

The model demonstrates high localized prediction confidence (0.80 to 1.0) for nearby pedestrians, domestic animals, and distinct vehicles under standard viewing conditions. Challenges occur with small, distant traffic indicators, heavily occluded elements, or poorly illuminated night targets.

VII. Conclusion and Future Work

This work details the implementation of a real-time object detection pipeline for autonomous vehicles operating in challenging urban road domains using a custom-engineered YOLOv8m network. Using a split two-stage training methodology, the fine-tuned model achieved a 0.75–0.78 mAP@0.5 accuracy profile while supporting inference throughput suitable for real-world driving safety systems.

Future research will focus on improving processing speed on edge hardware via structural layer optimization.

A. Integration of Depthwise Separable Convolutions

Standard 3×3 convolutions applied across input channels C_{in} to generate output channels C_{out} incur high computational costs:

$$\text{FLOPs}_{\text{Standard}} = H \times W \times C_{in} \times C_{out} \times 3 \times 3 \qquad (3)$$

By replacing standard convolution layers with **Depthwise Separable Convolutions** [7], this process is factorized into two separate steps: a spatial Depthwise Convolution followed by a Pointwise channel-mixing transformation. This reduces computational cost to:

$$\text{FLOPs}_{\text{Separable}} = H \times W \times C_{in} \times (9 + C_{out}) \qquad (4)$$

Integrating depthwise separable layers into the deeper C2f modules of the YOLOv8m backbone network can reduce parameter counts by 30% to 50%, enabling higher frame rates on low-power edge compute units.

B. Other Future Research Areas

Additional improvements include collecting real-world driving data under extreme weather anomalies (e.g., severe

dust storms, heavy downpours), exploring newer YOLOv11 attention-based network designs [6], and using structural weight pruning and INT8 model quantization to optimize inference efficiency on embedded target nodes.

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